Building and evaluation of a PBPK model for Desipramine in adults

Version	**evaluation-OSP12.2**
based on *Model Snapshot* and *Evaluation Plan*	https://github.com/Open-Systems-Pharmacology/Desipramine-Model/releases/tag/vevaluation
OSP Version	12.2
Qualification Framework Version	3.5

This evaluation report and the corresponding PK-Sim project file are filed at:

https://github.com/Open-Systems-Pharmacology/OSP-PBPK-Model-Library/

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1 Introduction

Desipramine is a tricyclic antidepressant and a sensitive CYP2D6 substrate. It is used as a probe substrate in clinical drug-drug interaction studies and is suitable for evaluating CYP2D6 phenotype- and activity-dependent changes in exposure.

This desipramine model is intended to describe plasma concentration-time profiles of desipramine and 2-hydroxydesipramine after intravenous and oral administration. It supports CYP2D6 drug-gene and drug-drug-gene interaction simulations in the CYP2D6 network described by Rüdesheim 2025.

The model was developed using physicochemical, *in vitro*, and clinical pharmacokinetic information for desipramine and 2-hydroxydesipramine. Clinical data included intravenous and oral administration, extensive-, poor-, normal-, and higher-activity CYP2D6 groups, and interacting-drug studies using desipramine as a CYP2D6 victim.

The presented model includes the following features:

- desipramine and 2-hydroxydesipramine as parent and metabolite compounds,
- CYP2D6-mediated 2-hydroxylation with activity-score-dependent k_{cat} values,
- residual hepatic clearance processes,
- renal filtration with a GFR fraction of 1,
- intravenous and oral applications used for clinical model building and verification.

2 Methods

2.1 Modeling strategy

The general concept of building a PBPK model has previously been described by Kuepfer et al. (Kuepfer 2016). Relevant information on anthropometric and physiological parameters in adults was gathered from the literature and incorporated into PK-Sim as default values for adult simulations (Willmann 2007).

The applied activity and variability of plasma proteins and active processes integrated into PK-Sim are described in the publicly available PK-Sim Ontogeny Database or otherwise referenced for the specific process.

The desipramine model was developed as a parent-metabolite PBPK model with desipramine and 2-hydroxydesipramine. It was included in the comprehensive CYP2D6 drug-drug-gene interaction network by Rüdesheim 2025. The model was built using intravenous and oral desipramine data to inform distribution, absorption, CYP2D6-mediated metabolism, and residual elimination.

Clinical studies used for model building covered intravenous and oral desipramine administration in CYP2D6 poor-, extensive-, and fast-metabolizer groups. Verification simulations included independent oral studies, activity-score groups, and CYP2D6 inhibitor interaction scenarios. The model therefore separates parent desipramine performance from 2-hydroxydesipramine performance where metabolite data are available.

The intravenous data support distribution and systemic clearance assumptions without confounding by oral absorption. Oral studies then support absorption and first-pass metabolism. This split is important for desipramine because the same model must describe both parent exposure and metabolite formation across CYP2D6 function groups.

The structural model contains CYP2D6-mediated 2-hydroxylation, residual hepatic clearance, passive renal filtration, and oral absorption. CYP2D6 poor-metabolizer activity is represented by zero CYP2D6 k_{cat}, while normal and higher activity-score groups use optimized CYP2D6 k_{cat} values.

The evaluated applications include intravenous infusion or bolus dosing and oral dosing. The major proteins and processes represented explicitly are CYP2D6, plasma protein binding, residual hepatic clearance, and passive renal filtration. Inhibition simulations are interpreted through the CYP2D6 pathway because desipramine is a sensitive CYP2D6 substrate.

The evaluation therefore uses parent concentration-time profiles as the primary readout and metabolite profiles where available as supporting evidence for the CYP2D6 formation pathway. Model-building and verification profiles are separated to preserve the logic of fitting structural parameters first and then evaluating independent clinical scenarios.

Details about input data are provided in Section 2.2. Details about the structural model and assumptions are provided in Section 2.3.

2.2 Data used

2.2.1 In vitro and physicochemical data

The table below summarizes the drug-dependent inputs documented for the desipramine model.

Parameter	Unit	Value	Source	Description
Desipramine MW	g/mol	266.4	PubChem 2019	Parent compound size used in concentration conversions.
Desipramine pK_a	-	2.84, 10.02	ChemAxon 2023; DrugBank 2006	Ionization constants.
Desipramine logP	-	3.52	Optimized; ChemAxon 2023; DrugBank 2006	Distribution-sensitive parent parameter.
Desipramine f_u	%	14	Watanabe 2018	Plasma binding input.
CYP2D6 K_m, desipramine to 2-hydroxydesipramine	µmol/L	0.73	Ball 1997; Austin 2002	CYP2D6 affinity held constant across phenotypes.
CYP2D6 k_{cat}, normal metabolizers	1/min	5.03	Optimized	CYP2D6 activity-score-dependent metabolic capacity.
CYP2D6 k_{cat}, AS = 2 and AS = 2.5	1/min	9.86, 28.88	Optimized	Higher activity-score metabolic capacity.
2-Hydroxydesipramine unspecific CL_{hep}	1/min	9.93	Optimized	Metabolite elimination.
GFR fraction	-	1.00	Assumed	Passive glomerular filtration fraction for parent and metabolite.

2.2.2 Clinical data

The evaluation uses 46 observed-data records for desipramine and 2-hydroxydesipramine in peripheral venous blood plasma. The evaluation plan assigns 12 simulations to model building and 8 simulations to model verification.

Model-building clinical data:

Publication	Arm / Treatment / Information used for model building
Brøsen 1988	Plasma PK profiles in adults after intravenous administration of 50 mg desipramine hydrochloride with CYP2D6 poor-, extensive-, and fast-metabolizer status.
Spina 1987	Plasma PK profiles in adults after oral, single-dose administration of 25 mg desipramine hydrochloride with CYP2D6 poor- and extensive-metabolizer status.
Aarnoutse 2005	Plasma PK profiles in adults after oral, single-dose administration of 50 mg desipramine hydrochloride with CYP2D6 extensive-metabolizer status.
Boni 2009	Plasma PK profiles in adults after oral, single-dose administration of 50 mg desipramine hydrochloride with desipramine and 2-hydroxydesipramine measurements.
Harris 2007	Plasma PK profiles in adults after oral, single-dose administration of 50 mg desipramine hydrochloride with CYP2D6 extensive-metabolizer status.
Madani 2002	Plasma PK profiles in adults after oral, single-dose administration of 50 mg desipramine hydrochloride with CYP2D6 extensive-metabolizer status.
Brøsen 1986	Plasma PK profiles in adults after oral, single-dose administration of 100 mg desipramine hydrochloride with CYP2D6 poor-, extensive-, and fast-metabolizer status.

Model-verification clinical data:

Publication	Arm / Treatment / Information used for model verification
Bergmann 2001	Plasma PK profiles in adults after oral, single-dose administration of 100 mg desipramine hydrochloride with CYP2D6 AS = 2 and AS = 2.5 status.
Hynes 2015	Plasma PK profiles in adults after oral, single-dose administration of 50 mg desipramine hydrochloride with CYP2D6 extensive-metabolizer status.
Bergstrom 1992	Plasma PK profiles in adults after oral, single-dose administration of 50 mg desipramine hydrochloride.
Nichols 2013	Plasma PK profiles in adults after oral, single-dose administration of 50 mg desipramine hydrochloride with CYP2D6 extensive-metabolizer status.
Patroneva 2008	Plasma PK profiles in adults after oral, single-dose administration of 50 mg desipramine hydrochloride with desipramine and 2-hydroxydesipramine measurements.
Skinner 2003	Plasma PK profiles in adults after oral, single-dose administration of 50 mg desipramine hydrochloride with CYP2D6 extensive-metabolizer status.
Spina 1995	Plasma PK profiles in adults after oral, single-dose administration of 100 mg desipramine hydrochloride with CYP2D6 extensive-metabolizer status.

2.3 Model parameters and assumptions

2.3.1 Absorption

The model includes intravenous and oral desipramine applications. Intravenous simulations do not require an absorption process. Oral desipramine simulations are represented with a solution-type formulation and compound-specific intestinal permeability.

The `Specific intestinal permeability` for desipramine was optimized using oral concentration-time data. The 2-hydroxydesipramine intestinal permeability was calculated and is relevant for metabolite simulations where the compound is present as a separate building block.

The intravenous simulations provide information for systemic disposition without absorption. Oral simulations use the same systemic disposition assumptions and add the intestinal permeability term. This allows the oral absorption parameter to be interpreted separately from the CYP2D6 and residual clearance parameters.

No phenotype-specific absorption parameter was introduced. Differences between CYP2D6 groups and inhibitor scenarios are represented through metabolism, while the same oral absorption assumptions are applied across adult simulations.

2.3.2 Distribution

Desipramine plasma protein binding was represented by a fraction unbound of 14% as summarized in Section 2.2.1. Lipophilicity was optimized and is an important parameter for the parent volume of distribution.

Partition coefficients were calculated with the Rodgers and Rowland method. Because the model includes both parent and metabolite concentration-time profiles, distribution assumptions affect the interpretation of desipramine and 2-hydroxydesipramine profiles separately.

The model uses compound-specific distribution properties for desipramine and 2-hydroxydesipramine. Parent distribution is particularly important because desipramine is extensively distributed and the concentration-time profile after intravenous administration constrains this part of the model. Metabolite distribution is interpreted more cautiously because metabolite profiles are less comprehensive than parent profiles.

Distribution was kept independent of CYP2D6 status. CYP2D6 poor-, extensive-, and higher-activity groups differ through formation clearance, while the same adult physiological and binding assumptions are applied.

2.3.3 Metabolism and Elimination

One explicit metabolic conversion and residual elimination pathways are represented in the model.

- CYP2D6

Desipramine is converted to 2-hydroxydesipramine by CYP2D6. CYP2D6 K_m is held constant across CYP2D6 groups. Poor-metabolizer k_{cat} is set to zero, while normal and higher activity-score k_{cat} values were optimized.

This structure assigns CYP2D6 phenotype and activity-score effects to catalytic capacity rather than to affinity. It also makes desipramine suitable for evaluating CYP2D6 inhibitor scenarios because reduced

CYP2D6 activity should increase parent exposure and reduce metabolite formation.

- Residual elimination

The model includes residual hepatic clearance for desipramine and unspecific hepatic clearance for 2-hydroxydesipramine. Both parent and metabolite include passive renal filtration with a `GFR fraction` of 1.

Residual hepatic clearance accounts for desipramine elimination not explicitly assigned to CYP2D6 2-hydroxylation. The metabolite unspecific hepatic clearance represents downstream elimination of 2-hydroxydesipramine. Renal filtration is implemented as a passive process using adult renal physiology and the fraction unbound of the respective compound.

2.3.4 Automated Parameter Identification

The following parameters were optimized by fitting the model to the data:

Model Parameter
`Lipophilicity`
`Specific intestinal permeability`
CYP2D6 k_{cat} values
desipramine residual hepatic clearance
2-hydroxydesipramine unspecific CL_{hep}

The optimized parameters were selected to describe distribution, oral absorption, CYP2D6-mediated formation, and residual clearance. Parameters with direct physicochemical or plasma-binding sources were kept fixed. The separation of optimized absorption and clearance terms is necessary because both oral and intravenous studies contribute to the evaluation.

3 Results and Discussion

The PBPK model for desipramine was developed and evaluated with clinical pharmacokinetic data after intravenous and oral administration. The evaluation covers desipramine and 2-hydroxydesipramine plasma concentration-time profiles, CYP2D6 poor-, extensive-, normal-, and higher-activity groups, and interaction settings in which desipramine acts as a sensitive CYP2D6 victim drug.

The model-building data informed systemic disposition, oral absorption, CYP2D6-mediated 2-hydroxylation, residual hepatic clearance, and metabolite elimination. Intravenous and oral phenotype-stratified data from Brøsen 1988, Brøsen 1986, and Spina 1987 supported CYP2D6 phenotype-dependent clearance. Oral studies with parent and metabolite measurements, including Boni 2009, supported the parent-metabolite structure. Additional oral data from Aarnoutse 2005, Harris 2007, and Madani 2002 supported adult oral exposure after 50 mg dosing.

Verification used independent studies across CYP2D6 activity and interaction settings, including Bergmann 2001, Hynes 2015, Bergstrom 1992, Nichols 2013, Patroneva 2008, Skinner 2003, and Spina 1995. These data test whether the same model can describe adult oral desipramine exposure without study-specific refitting.

The model quantifies CYP2D6-mediated 2-hydroxylation, residual hepatic clearance of desipramine, unspecific hepatic clearance of 2-hydroxydesipramine, passive glomerular filtration, and enterohepatic circulation. CYP2D6 k_{cat} values represent the major phenotype and activity-score dependency. The metabolite pathway is important because adequate parent prediction alone would not demonstrate correct description of 2-hydroxydesipramine exposure.

The next sections show:

1. the final model input parameters for the building blocks: Section 3.1.
2. the overall goodness of fit: Section 3.2.
3. simulated vs. observed concentration-time profiles for the clinical studies used for model building and for model verification: Section 3.3.

The merged GOF diagnostic over the included concentration observations gives GMFE values of 1.28 for desipramine, 1.27 for 2-hydroxydesipramine, and 1.28 for all observations. The close agreement between parent and metabolite GMFE values is important because it indicates that the model is not fitting parent exposure at the expense of metabolite behavior.

The concentration-time profiles should be interpreted by dose route, analyte, and CYP2D6 activity group. Intravenous data mainly test systemic distribution and clearance. Oral data additionally test absorption and first-pass metabolism. Poor-metabolizer profiles are expected to show reduced CYP2D6-mediated 2-hydroxylation, while normal- and higher-activity groups test the activity-score-dependent k_{cat} values.

Rüdesheim 2025 used the desipramine model in a broader CYP2D6 DDGI network. In that setting, desipramine served as a sensitive CYP2D6 substrate for inhibition and genotype interaction scenarios. The present evaluation should therefore be read as a compound-level check of desipramine and 2-

hydroxydesipramine exposure, while DDGI performance depends additionally on perpetrator models and interaction constants.

The model is adequate for adult desipramine simulations within the represented intravenous and oral dose range and CYP2D6 activity groups. Extrapolation to unrepresented populations, strong P-gp effects, or tissue-specific pharmacodynamic endpoints should remain outside the evidence supported by this report.

3.1 Desipramine final input parameters

The compound parameter values of the final PBPK model are illustrated below.

Compound: Desipramine

Parameters

Name	Value	Value Origin	Alternative	Default
Solubility at reference pH	214.29 mg/l	Database-Other-chemicalize.com	Predicted	True
Reference pH	6.5	Database-Other-chemicalize.com	Predicted	True
Lipophilicity	3.5204587578 Log Units	Parameter Identification-Parameter Identification-Optimized	Optimized	True
Lipophilicity	3.896 Log Units	Database-Other-chemicalize.com	Predicted	False
Fraction unbound (plasma, reference value)	0.135	Publication-Other-Predicted, Watanabe et al. 2018	Predicted	True
Specific intestinal permeability (transcellular)	1.14E-05 cm/min	Parameter Identification-Parameter Identification-Optimized	Optimized	True
Is small molecule	Yes			
Molecular weight	266.388 g/mol	Database-Other-chemicalize.com		
Plasma protein binding partner	α1-acid glycoprotein			

Calculation methods

Name	Value
Partition coefficients	Rodgers and Rowland
Cellular permeabilities	PK-Sim Standard

Processes

Systemic Process: Glomerular Filtration-Assumption

Species: Human

Parameters

Name	Value	Value Origin
GFR fraction	1	Parameter Identification-Parameter Identification-Optimized

Systemic Process: Total Hepatic Clearance-Assumption

Species: Human

Parameters

Name	Value	Value Origin
Fraction unbound (experiment)	0.14	Unknown-Unknown
Lipophilicity (experiment)	3.52 Log Units	
Plasma clearance	0 ml/min/kg	
Specific clearance	0.5395490378 1/min	Parameter Identification-Parameter Identification-Optimized

Metabolizing Enzyme: CYP2D6-Ball et al. 1997 (2-Hydroxydesipramine)

Molecule: CYP2D6

Metabolite: 2-Hydroxydesipramine

Parameters

Name	Value	Value Origin
In vitro Vmax for liver microsomes	68 pmol/min/mg mic. protein	
Content of CYP proteins in liver microsomes	10 pmol/mg mic. protein	Unknown
Km	0.732 µmol/l	Publication-In Vitro-Ball et al. 1997, corrected for fu,inc 12% (Austin et al. 2002)
kcat	5.03 1/min	Parameter Identification-Parameter Identification-Optimized

Metabolizing Enzyme: CYP2D6-AS=0 (2-Hydroxydesipramine)

Molecule: CYP2D6

Metabolite: 2-Hydroxydesipramine

Parameters

Name	Value	Value Origin
In vitro Vmax for liver microsomes	0 pmol/min/mg mic. protein	
Content of CYP proteins in liver microsomes	10 pmol/mg mic. protein	Unknown
Km	0.73 µmol/l	Publication-In Vitro-Ball et al. 1997, corrected for fu,inc 12% (Austin et al. 2002)

Metabolizing Enzyme: CYP2D6-AS=2 (2-Hydroxydesipramine)

Molecule: CYP2D6

Metabolite: 2-Hydroxydesipramine

Parameters

Name	Value	Value Origin
In vitro Vmax for liver microsomes	0 pmol/min/mg mic. protein	
Content of CYP proteins in liver microsomes	10 pmol/mg mic. protein	Unknown
Km	0.732 µmol/l	Publication-In Vitro-Ball et al. 1997, corrected for fu,inc 12% (Austin et al. 2002)
kcat	9.86 1/min	Parameter Identification-Parameter Identification-Optimized

Metabolizing Enzyme: CYP2D6-AS=2.5 (2-Hydroxydesipramine)

Molecule: CYP2D6

Parameters

Name	Value	Value Origin
In vitro Vmax for liver microsomes	0 pmol/min/mg mic. protein	
Content of CYP proteins in liver microsomes	10 pmol/mg mic. protein	Unknown
Km	0.732 µmol/l	Publication-In Vitro-Ball et al. 1997, corrected for fu,inc 12% (Austin et al. 2002)
kcat	28.88 1/min	Parameter Identification-Parameter Identification-Optimized

Compound: 2-Hydroxydesipramine

Parameters

Name	Value	Value Origin	Alternative	Default
Solubility at reference pH	282.39 mg/l	Database-Other-chemicalize.com	Predicted	True
Reference pH	6.5	Database-Other-chemicalize.com	Predicted	True
Lipophilicity	2.33 Log Units	Parameter Identification-Parameter Identification-Optimized	Optimized	True
Lipophilicity	3.173 Log Units	Database-Other-chemicalize.com	Predicted	False
Fraction unbound (plasma, reference value)	0.12	Publication-Other-Predicted, Watanabe et al. 2018	Predicted	True
Is small molecule	Yes			
Molecular weight	282.38 g/mol	Database-Other-chemicalize.com		
Plasma protein binding partner	Unknown			

Calculation methods

Name	Value
Partition coefficients	Rodgers and Rowland
Cellular permeabilities	PK-Sim Standard

Processes

Systemic Process: Glomerular Filtration-Assumption

Species: Human

Parameters

Name	Value	Value Origin
GFR fraction	1	Other-Other-Assumption

Systemic Process: Total Hepatic Clearance-Assumption

Species: Human

Parameters

Name	Value	Value Origin
Fraction unbound (experiment)	0.12	
Lipophilicity (experiment)	2.53 Log Units	Unknown-Unknown
Plasma clearance	0 ml/min/kg	
Specific clearance	9.93 1/min	Parameter Identification-Parameter Identification-Optimized

3.2 Diagnostic plots

Below you find the goodness-of-fit visual diagnostic plots for the PBPK model performance of all data used presented in Section 2.2.2.

The first plot shows observed versus simulated plasma concentration, the second weighted residuals versus time.

3.2.1 Desipramine goodness-of-fit diagnostics

Below you find the goodness-of-fit visual diagnostic plots for desipramine plasma concentration data used in the model evaluation.

The first plot shows observed versus simulated plasma concentration, the second weighted residuals versus time.

Table 3-1: GMFE for Desipramine observed versus simulated plasma concentration-time data

Group	GMFE
Desipramine	1.28

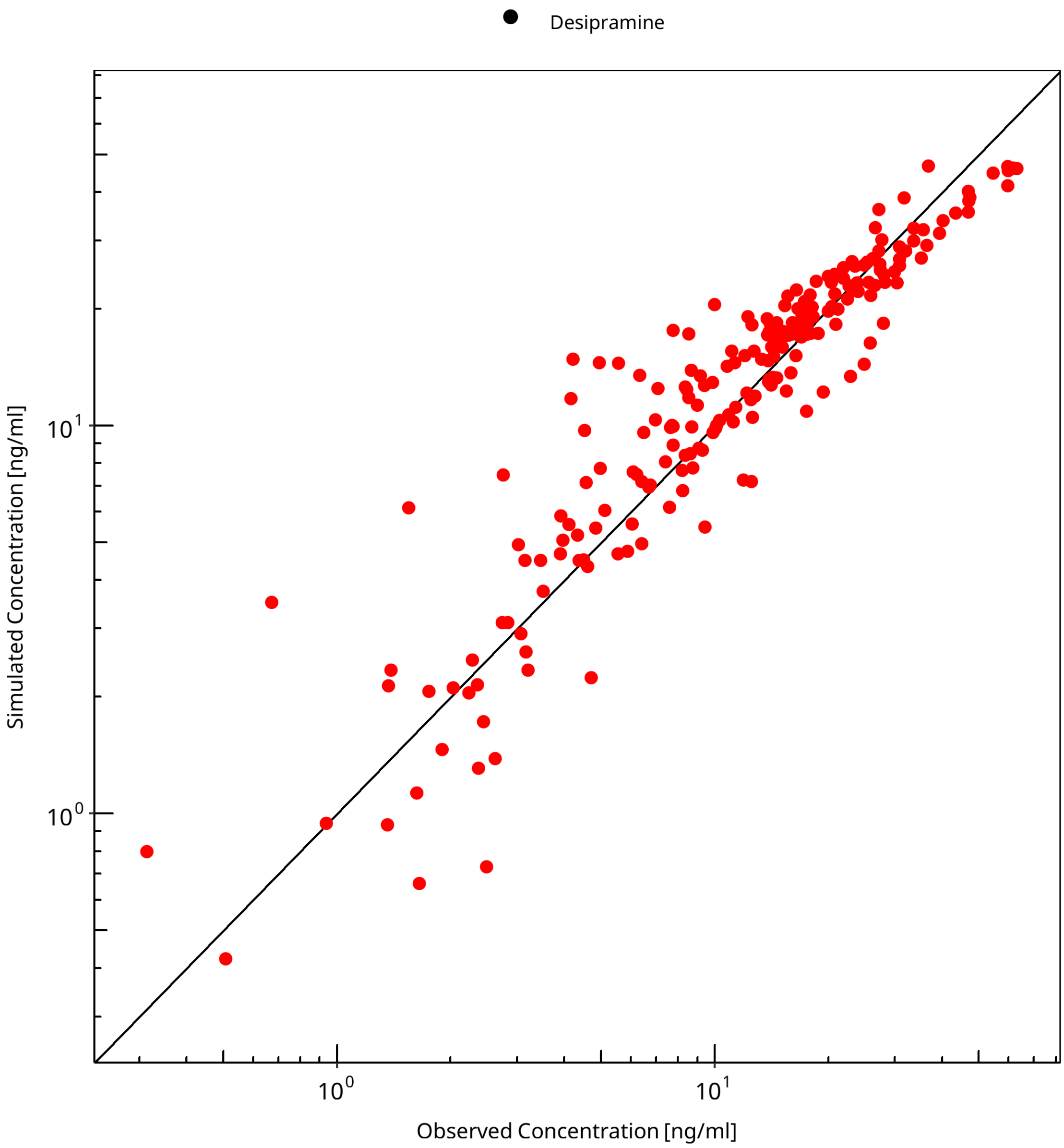

Figure 3-1: Desipramine observed versus simulated plasma concentration-time data

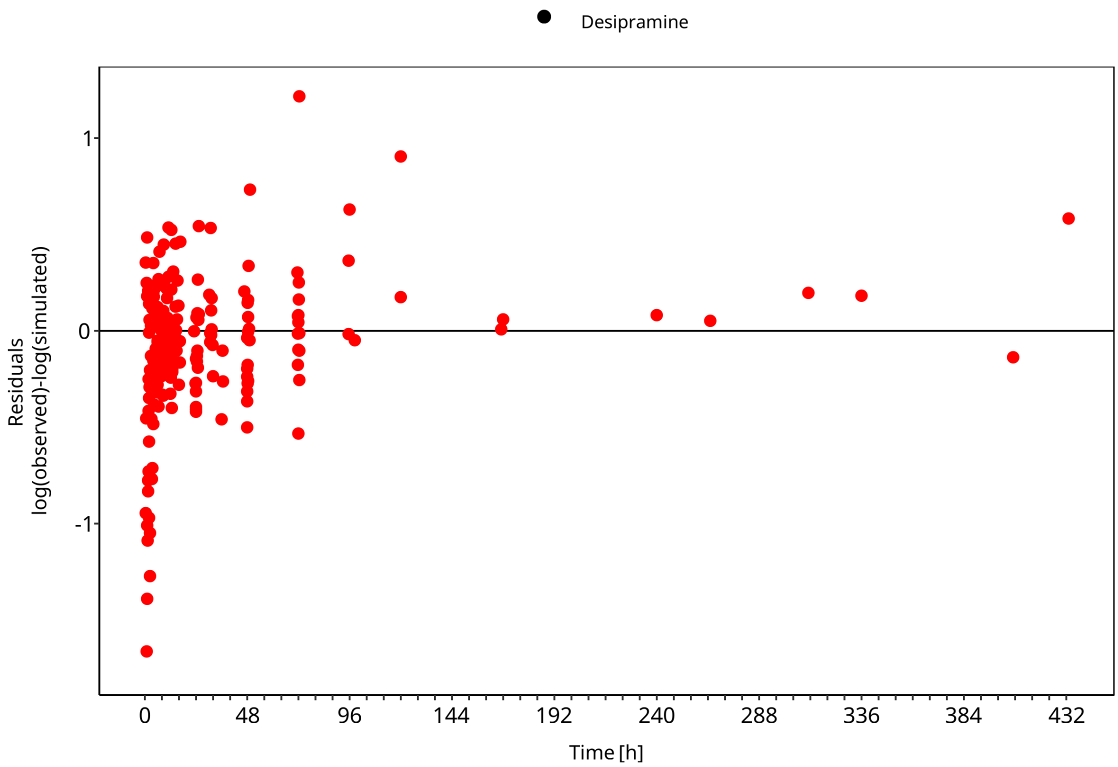

Figure 3-2: Desipramine observed versus simulated plasma concentration-time data

3.2.2 2-Hydroxydesipramine goodness-of-fit diagnostics

Below you find the goodness-of-fit visual diagnostic plots for 2-hydroxydesipramine plasma concentration data used in the model evaluation.

The first plot shows observed versus simulated plasma concentration, the second weighted residuals versus time.

Table 3-2: GMFE for 2-Hydroxydesipramine observed versus simulated plasma concentration-time data

Group	GMFE
2-Hydroxydesipramine	1.27

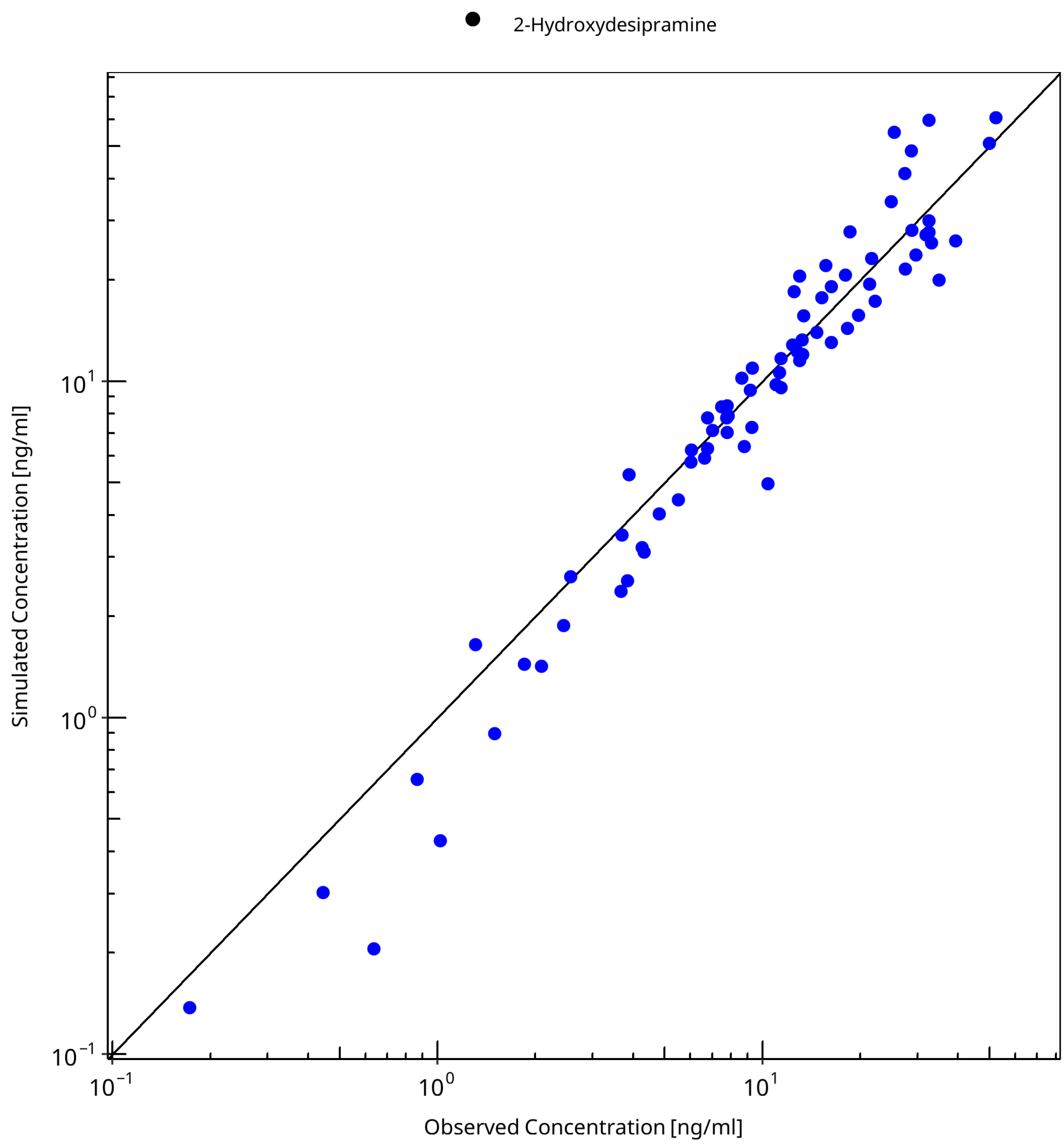

Figure 3-3: 2-Hydroxydesipramine observed versus simulated plasma concentration-time data

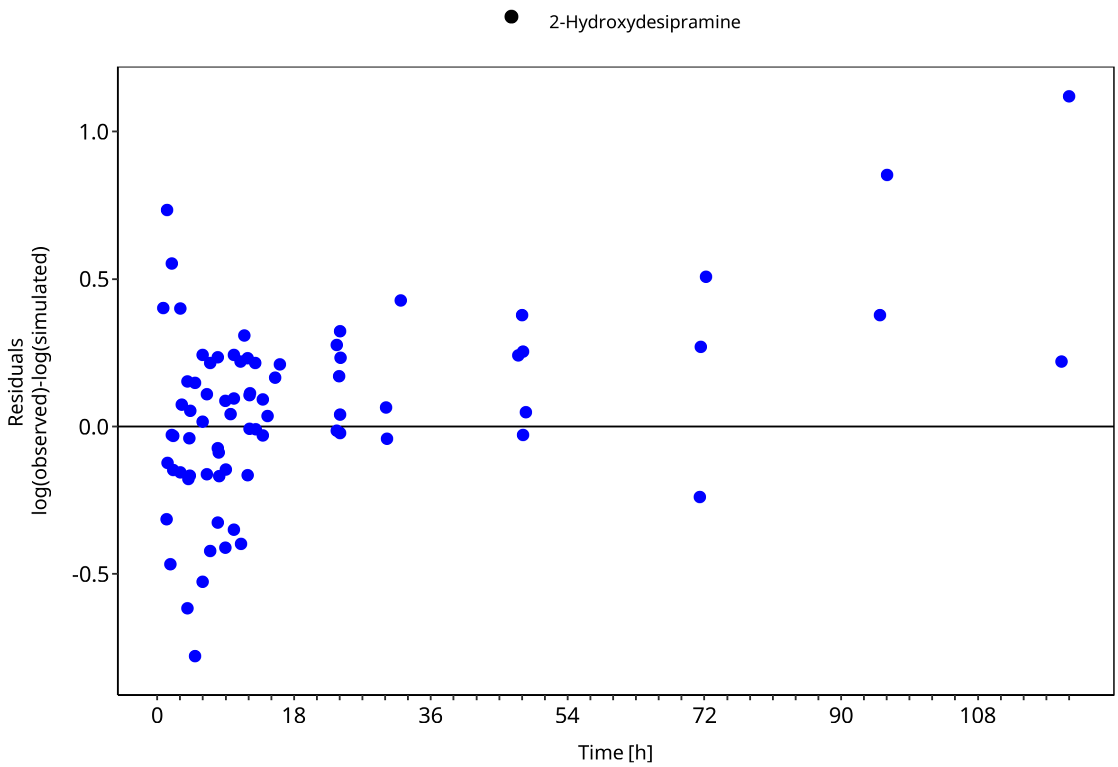

Figure 3-4: 2-Hydroxydesipramine observed versus simulated plasma concentration-time data

3.3 Concentration-time profiles

Simulated versus observed concentration-time profiles of all data listed in Section 2.2.2 are presented below.

3.3.1 Model Building

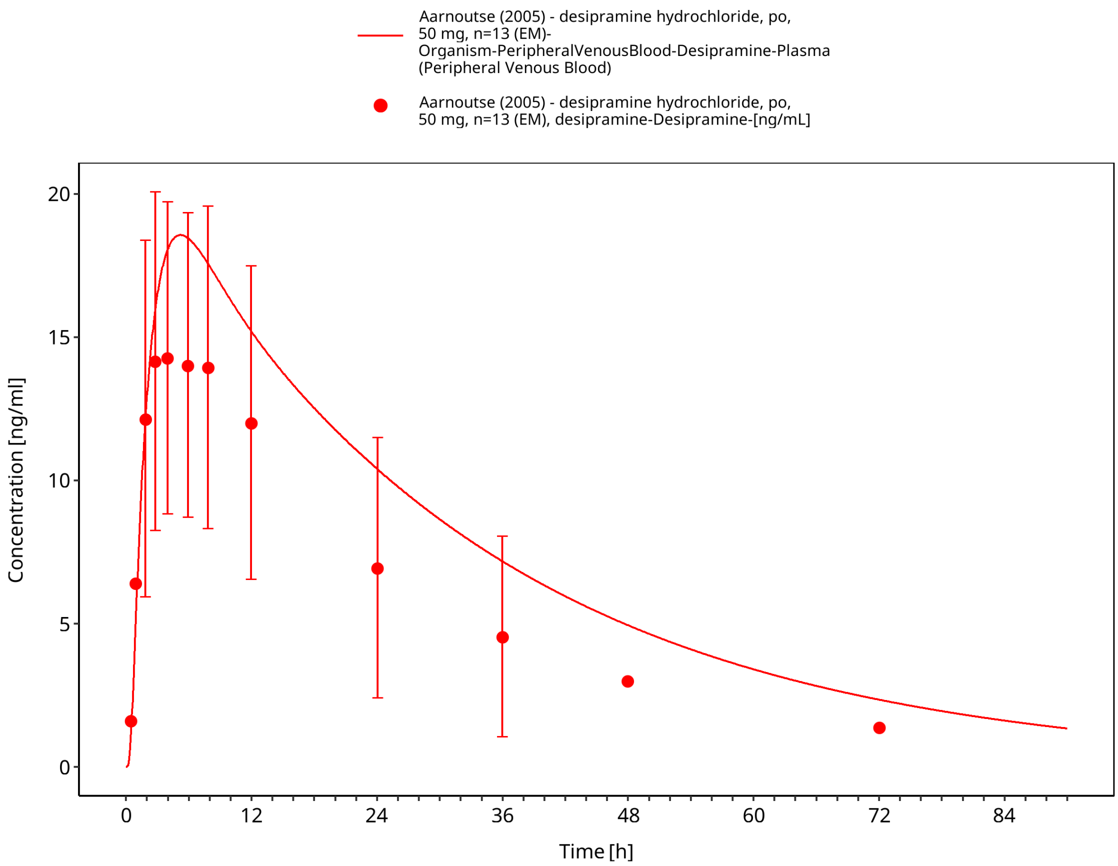

Figure 3-5: Time Profile Analysis

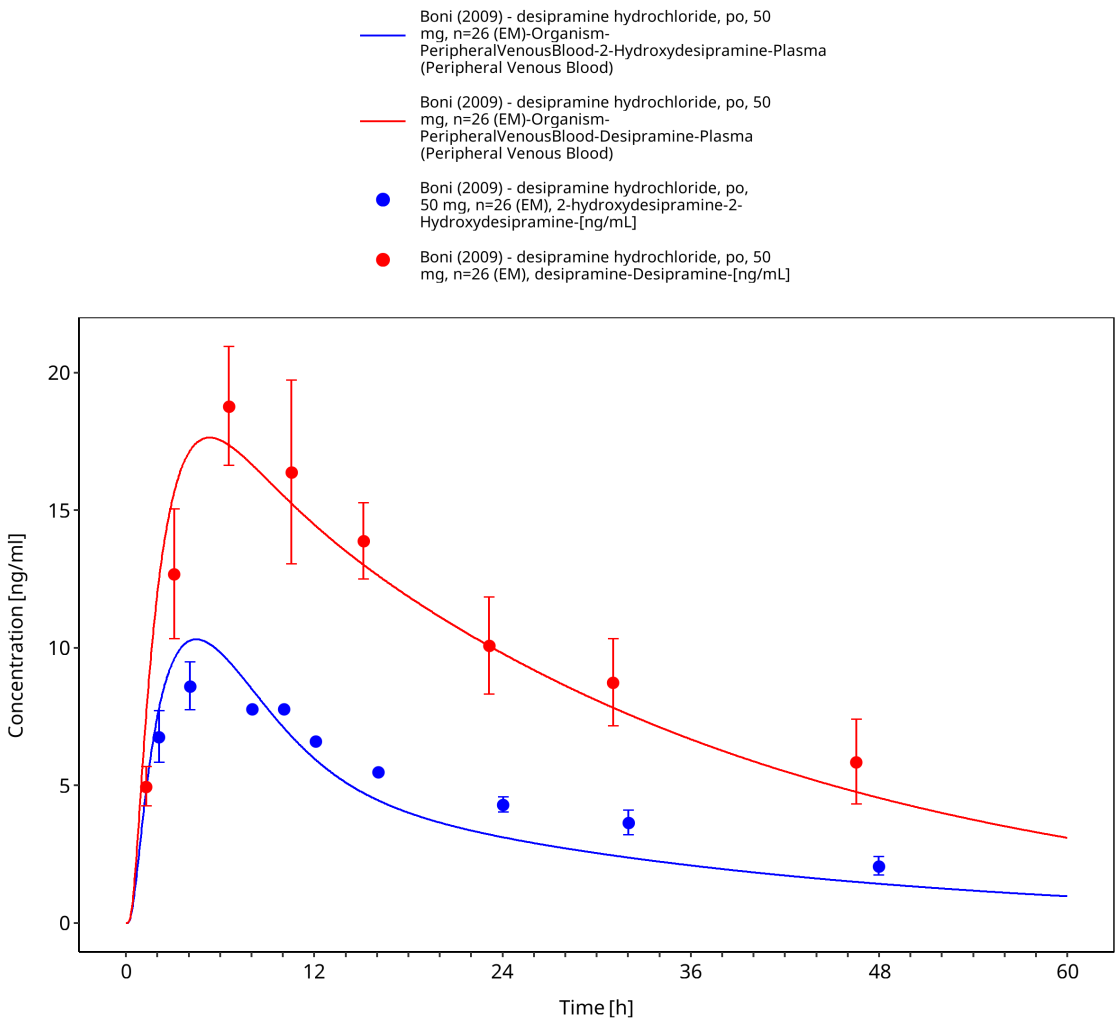

Figure 3-6: Time Profile Analysis

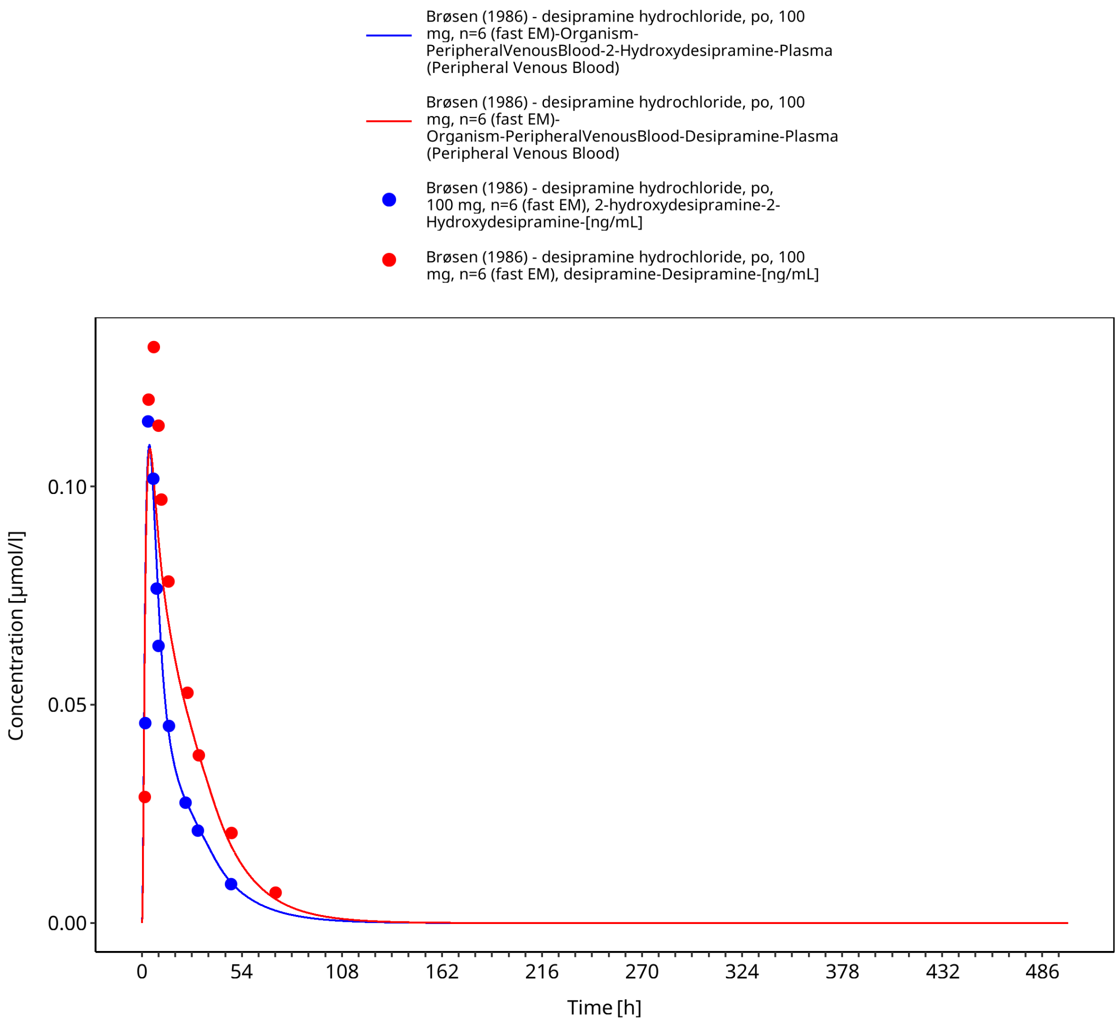

Figure 3-7: Time Profile Analysis

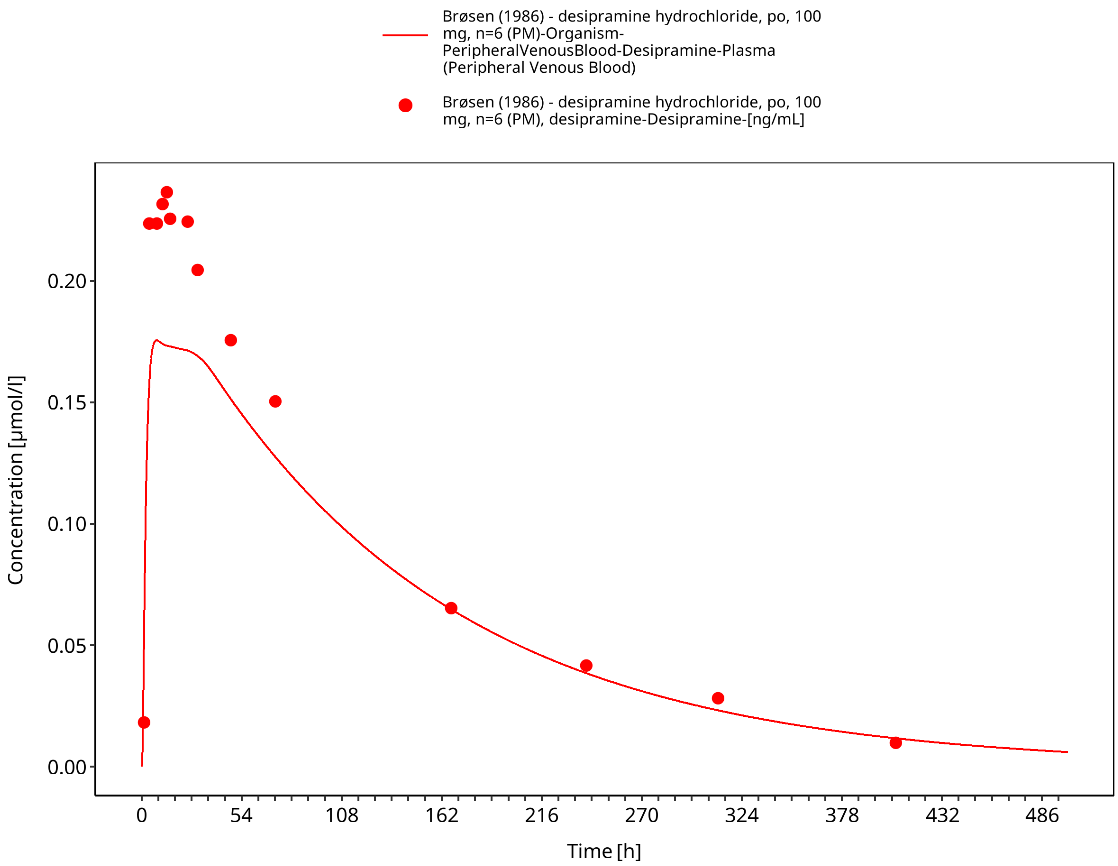

Figure 3-8: Time Profile Analysis

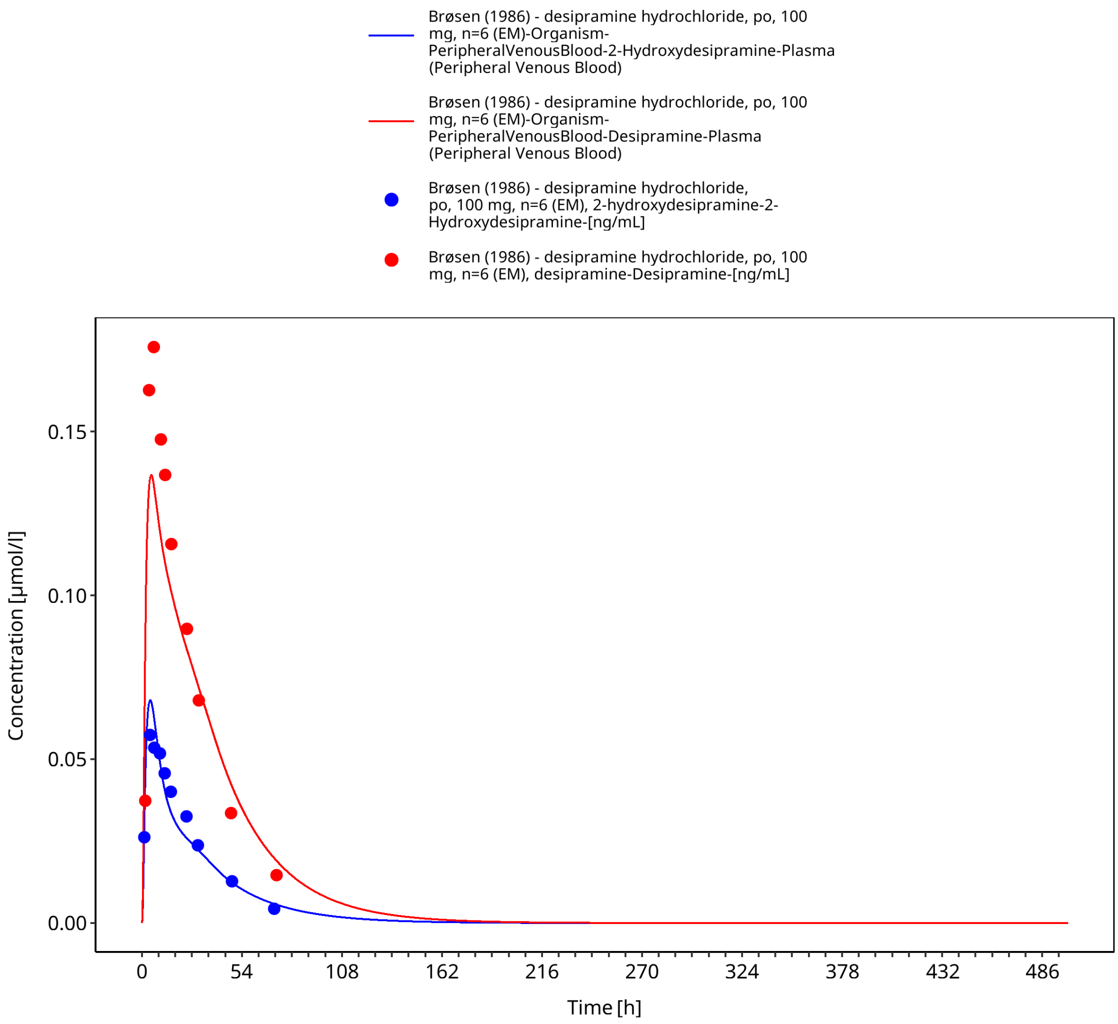

Figure 3-9: Time Profile Analysis

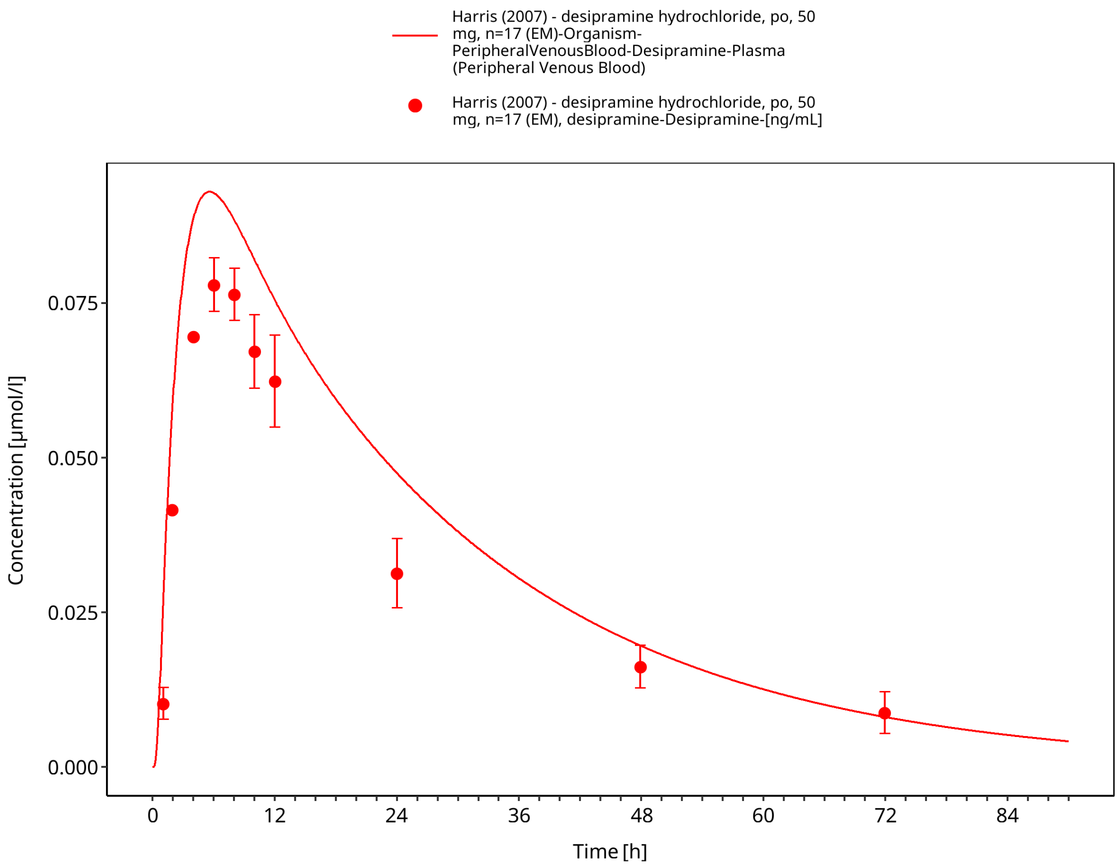

Figure 3-10: Time Profile Analysis

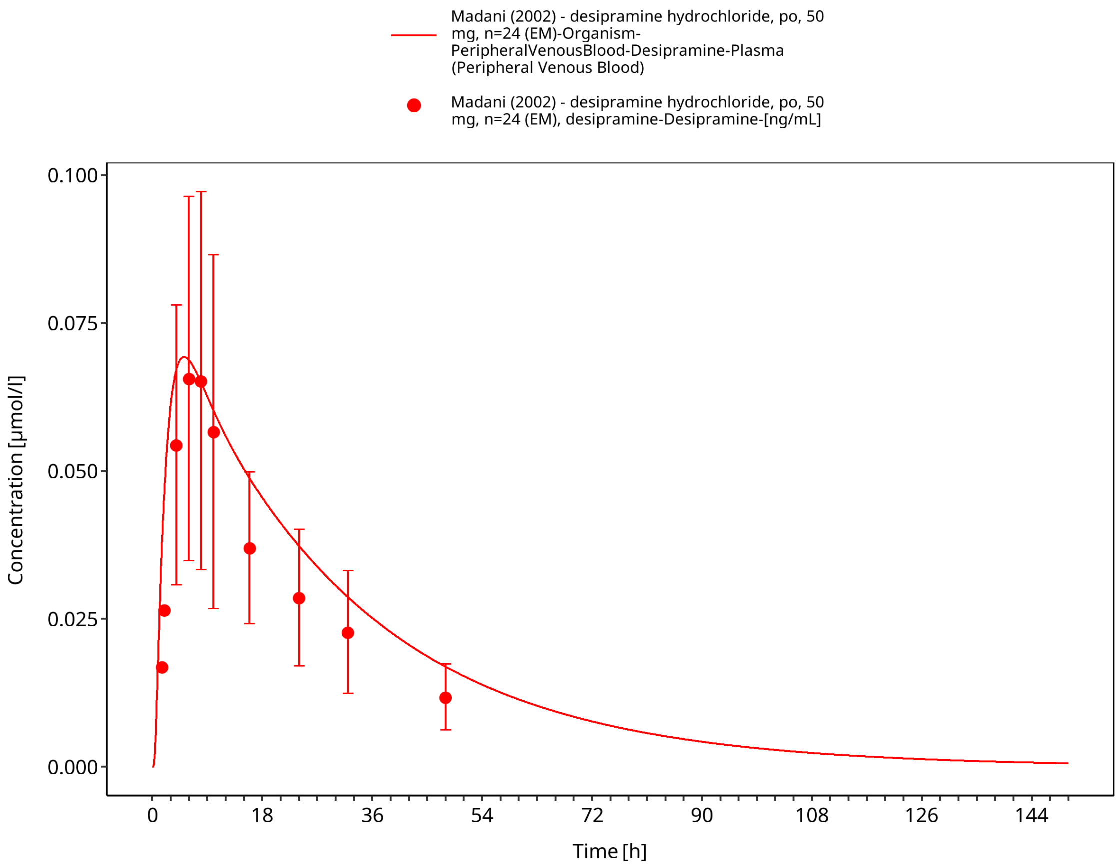

Figure 3-11: Time Profile Analysis

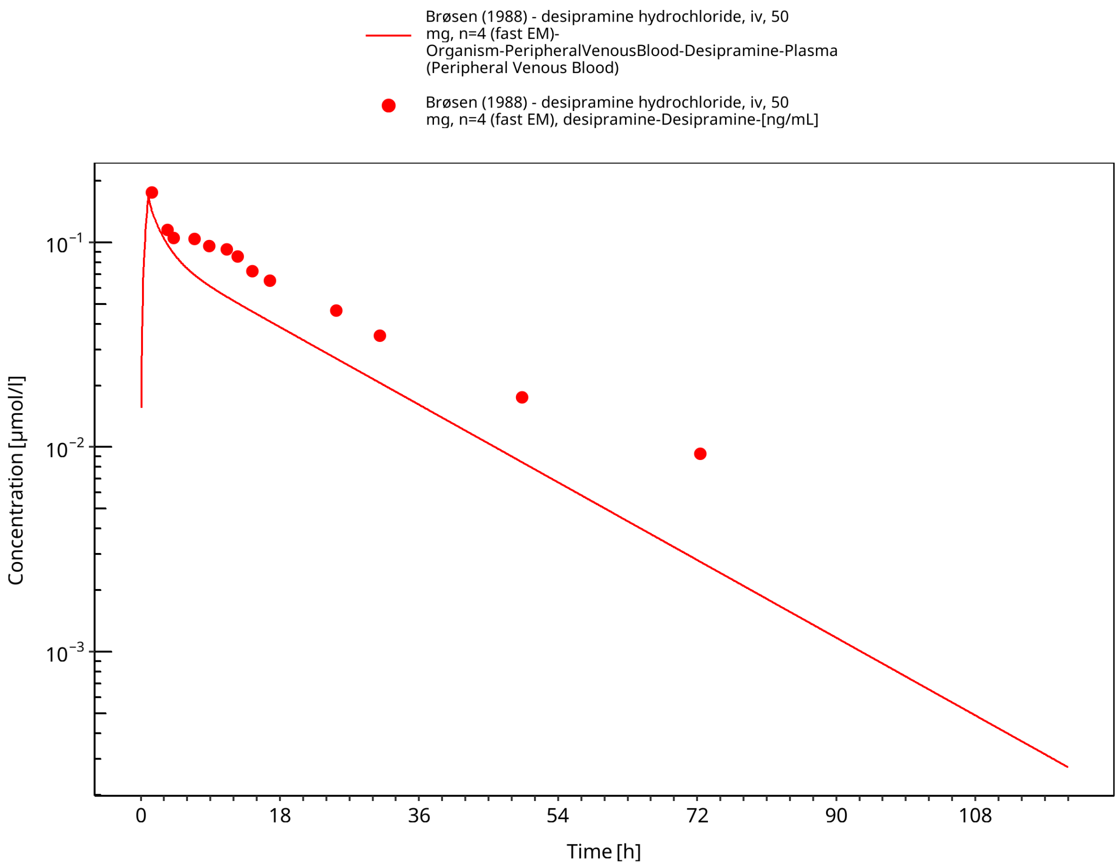

Figure 3-12: Time Profile Analysis

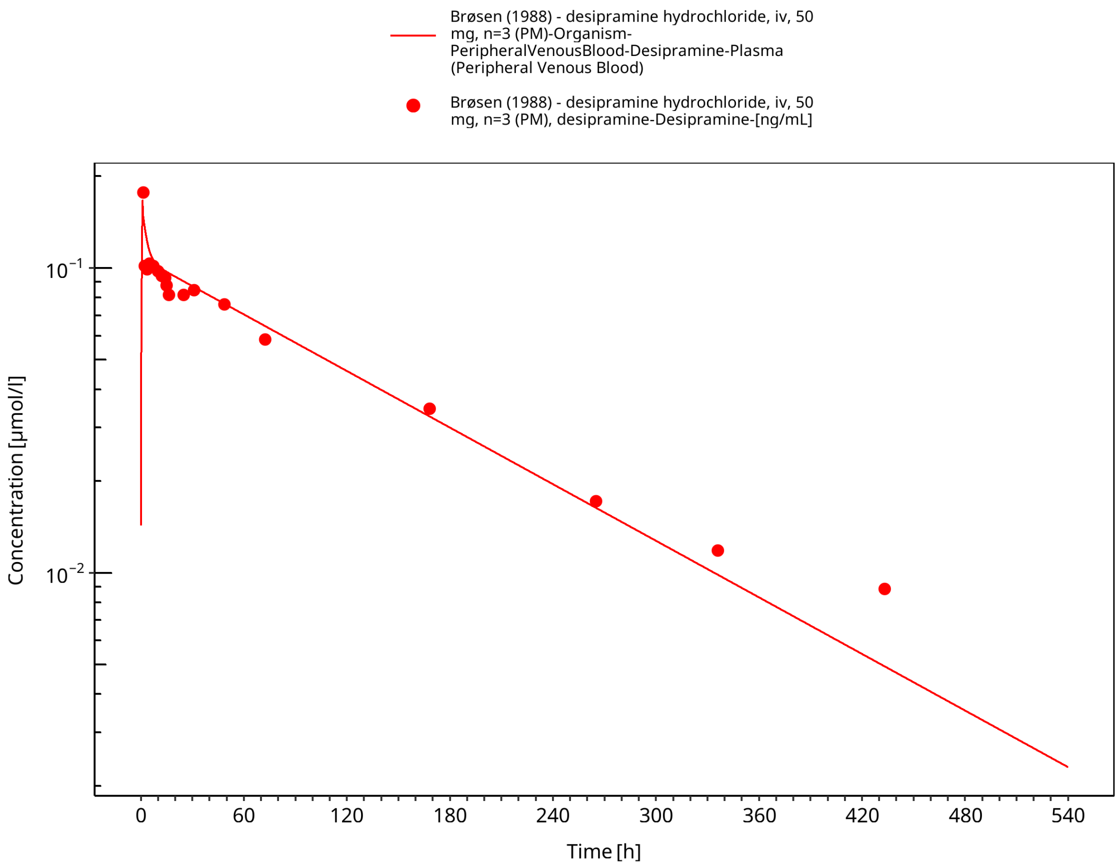

Figure 3-13: Time Profile Analysis

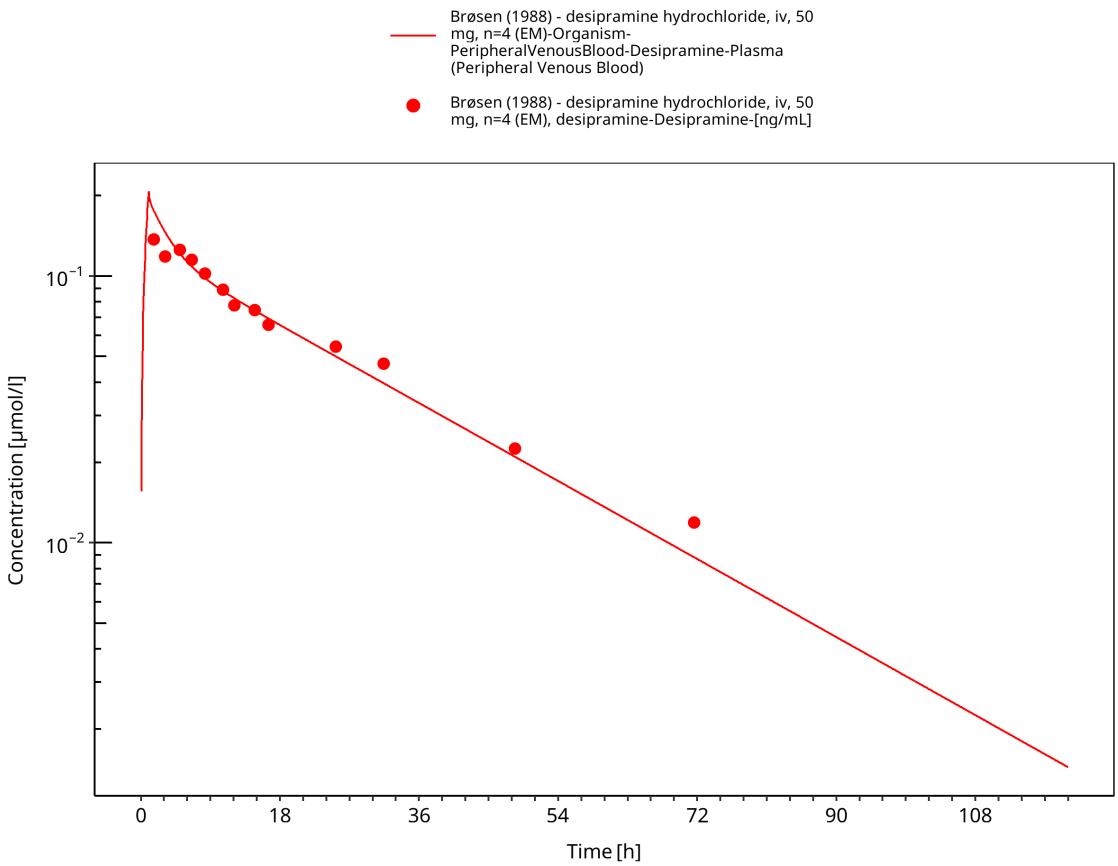

Figure 3-14: Time Profile Analysis

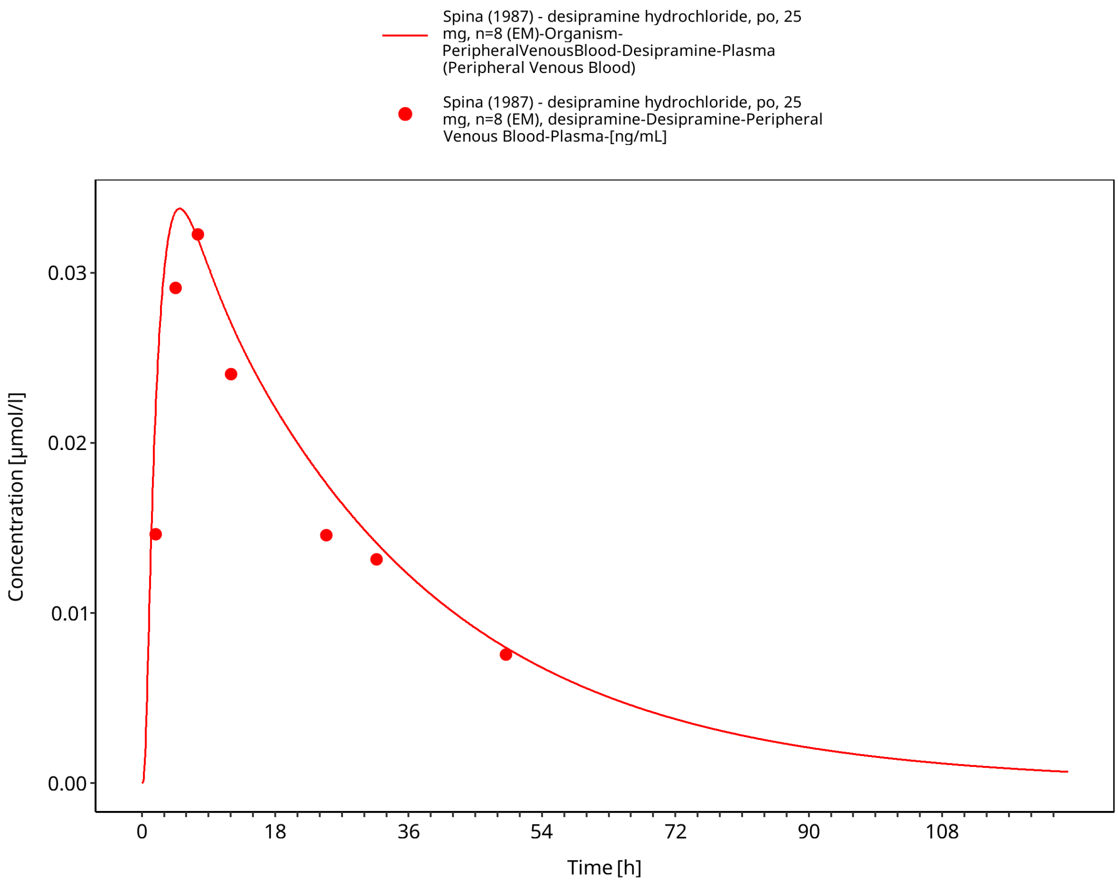

Figure 3-15: Time Profile Analysis

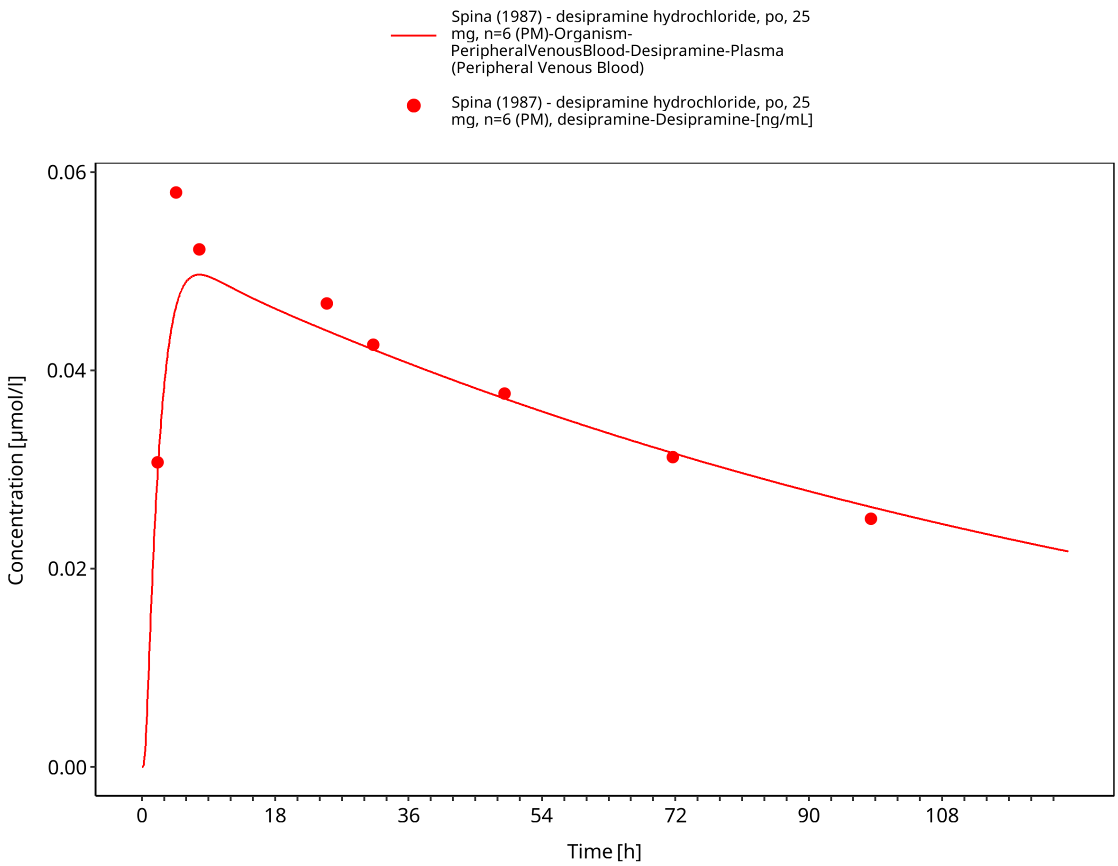

Figure 3-16: Time Profile Analysis

3.3.2 Model Verification

Simulated versus observed concentration-time profiles for model verification studies are presented below.

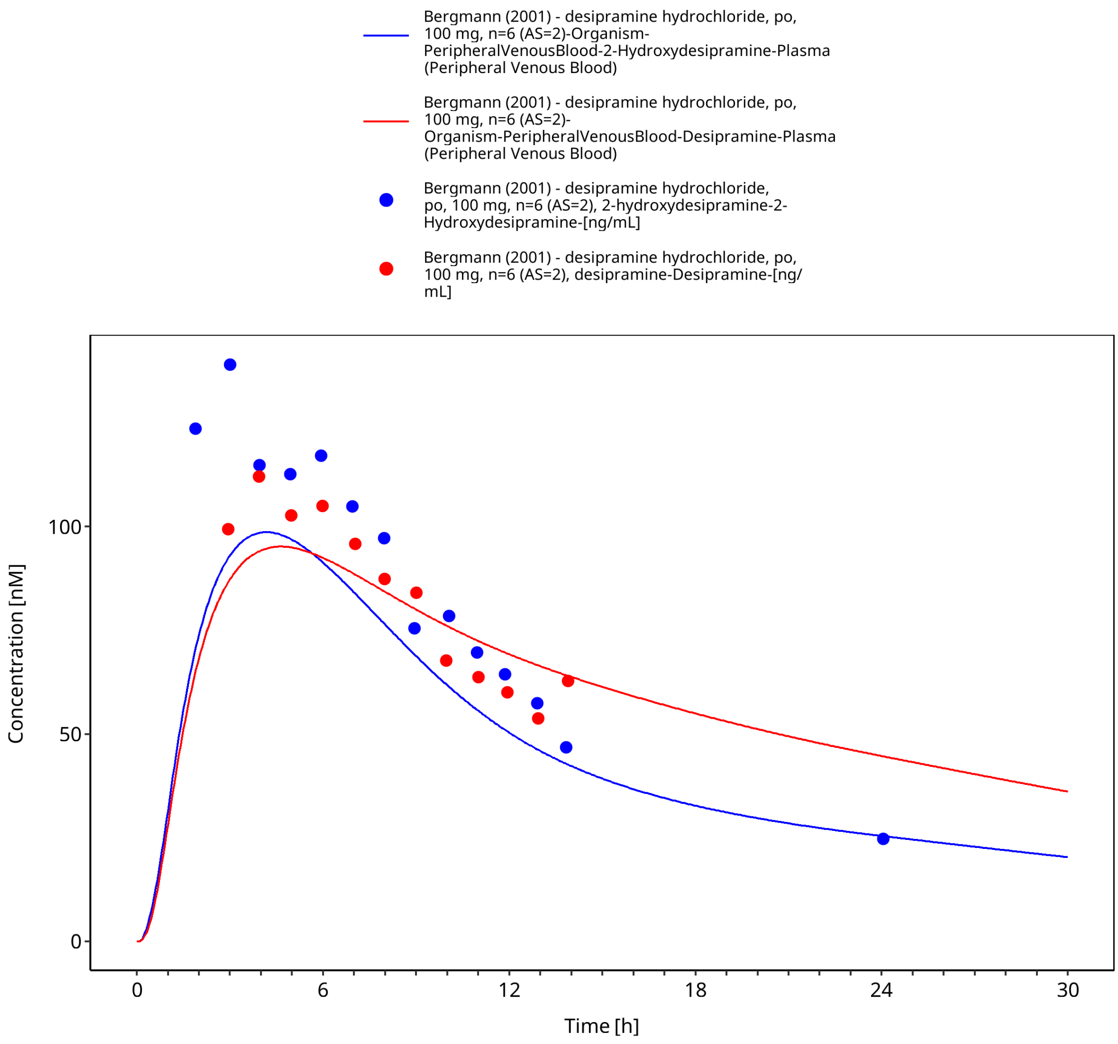

Figure 3-17: Time Profile Analysis

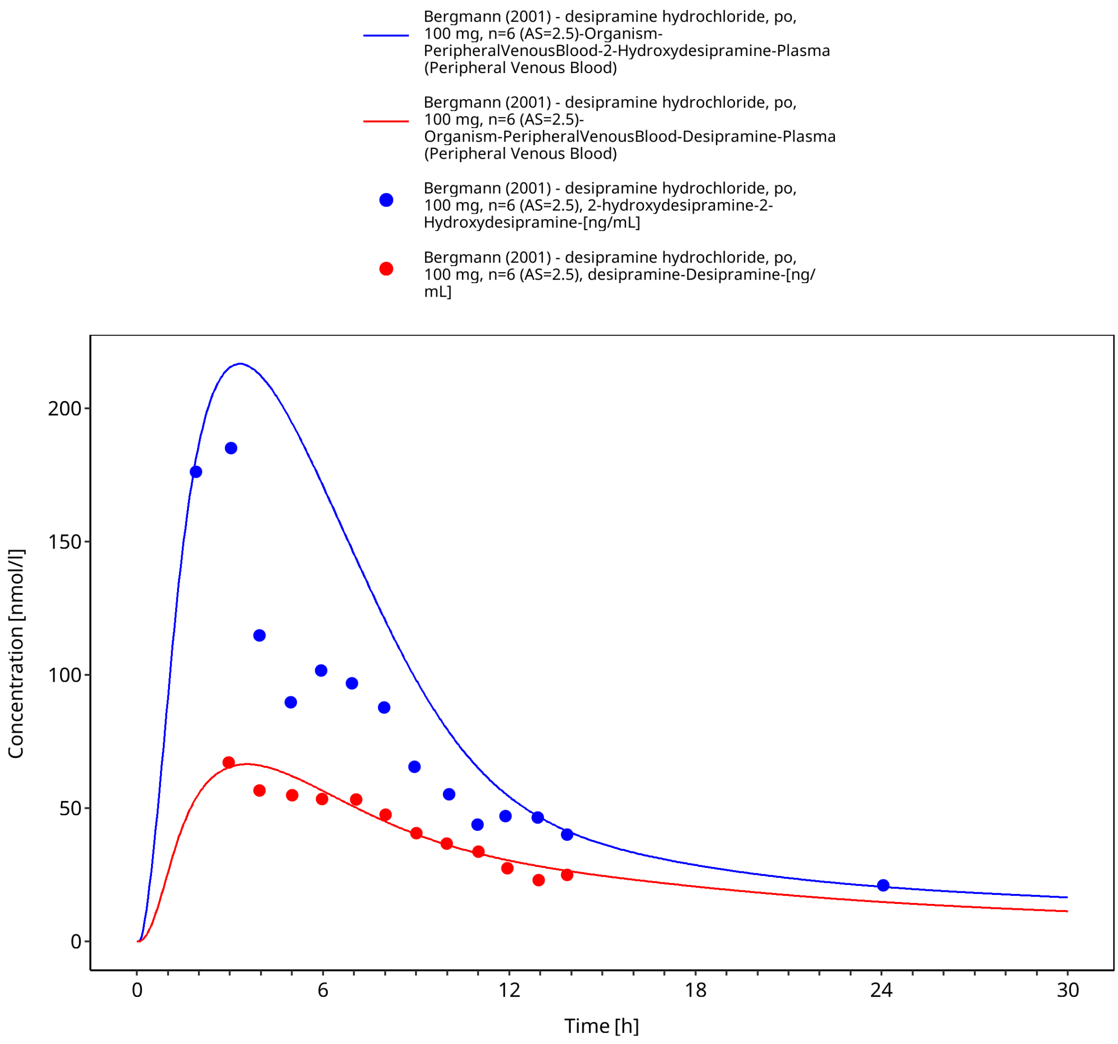

Figure 3-18: Time Profile Analysis

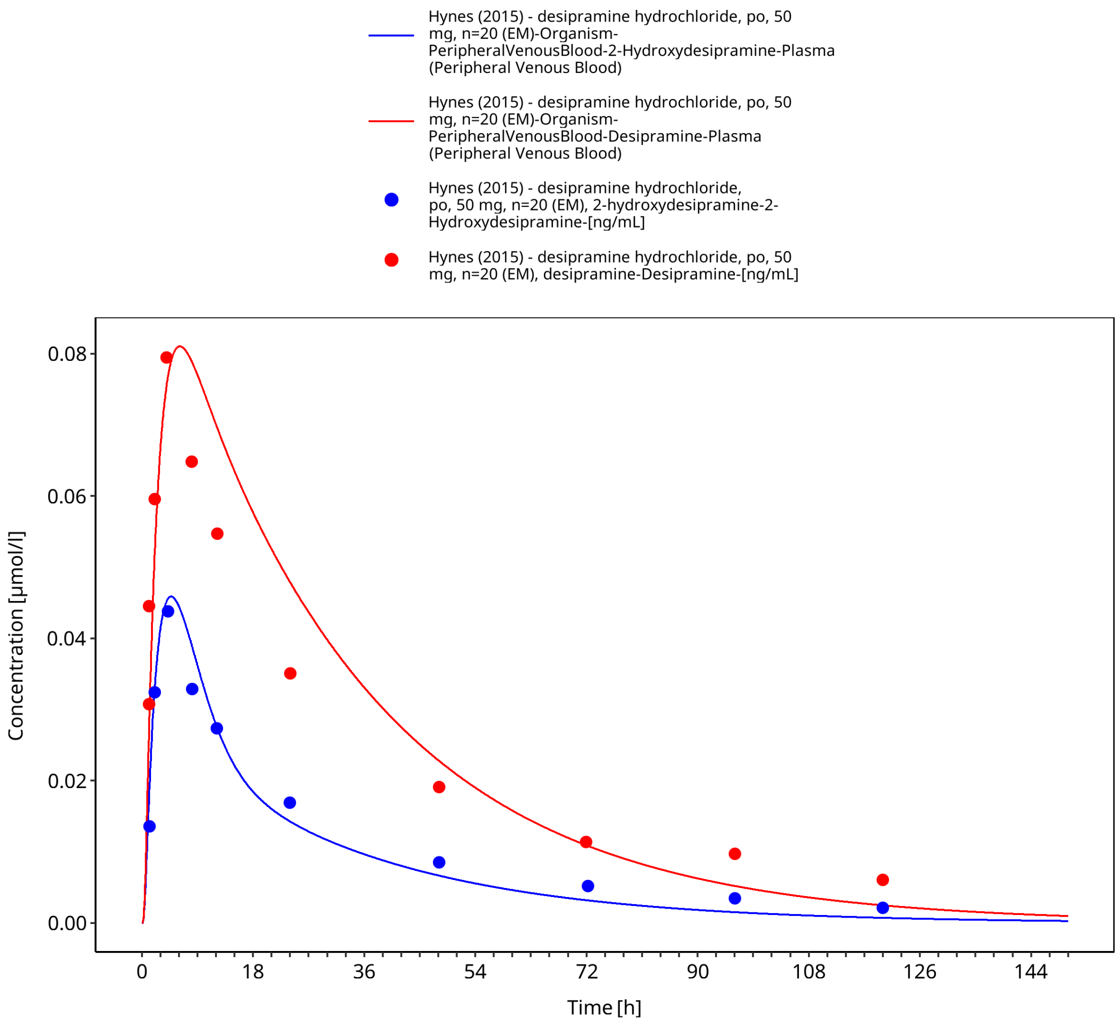

Figure 3-19: Time Profile Analysis

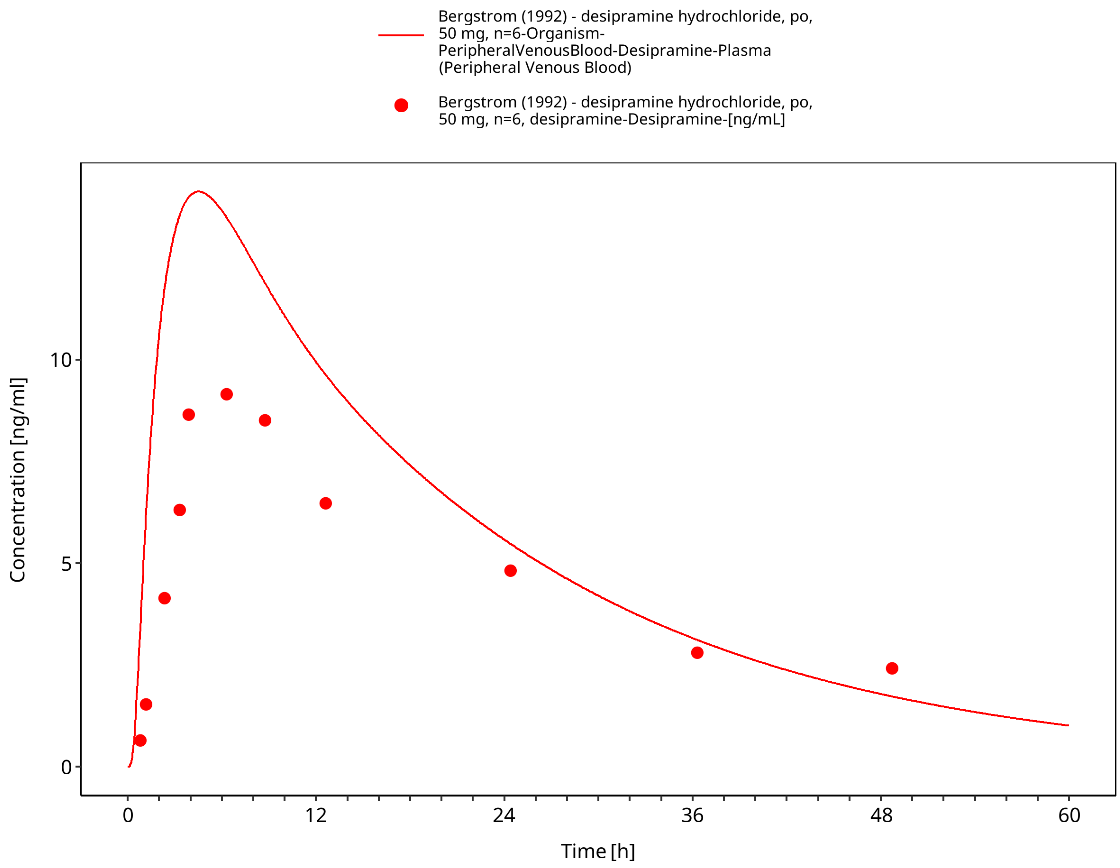

Figure 3-20: Time Profile Analysis

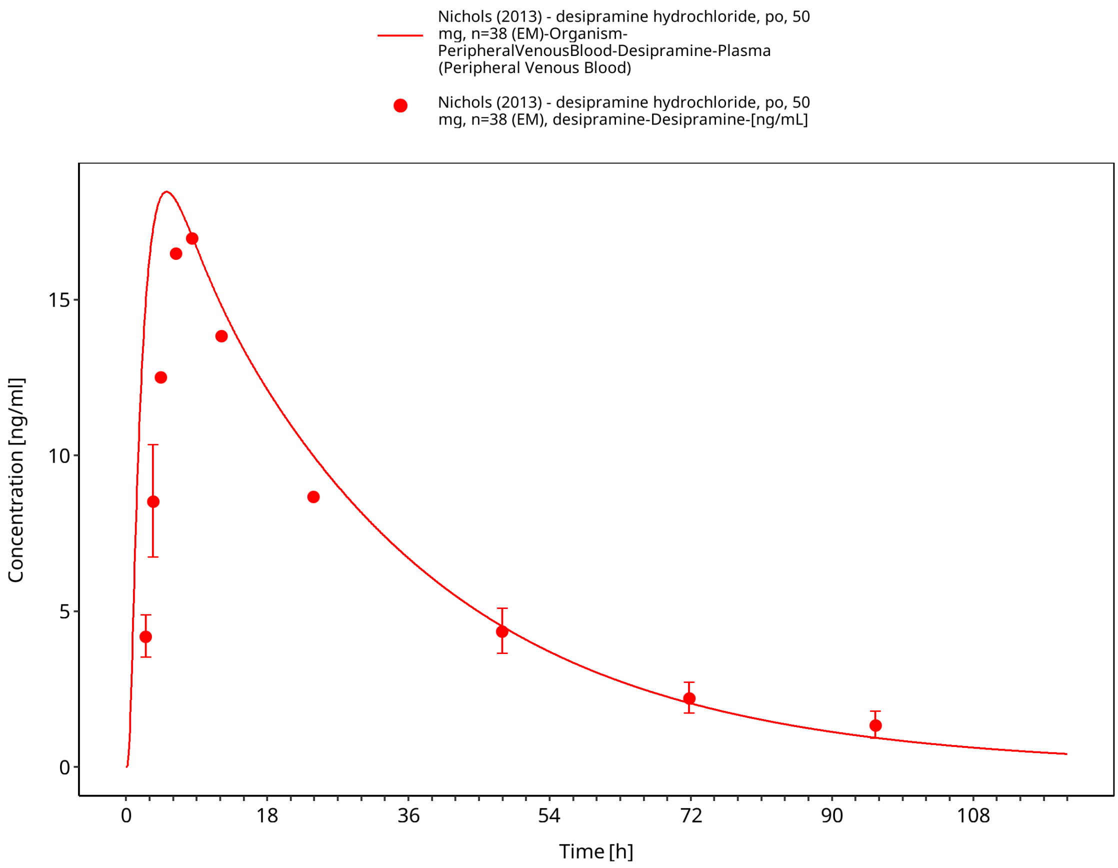

Figure 3-21: Time Profile Analysis

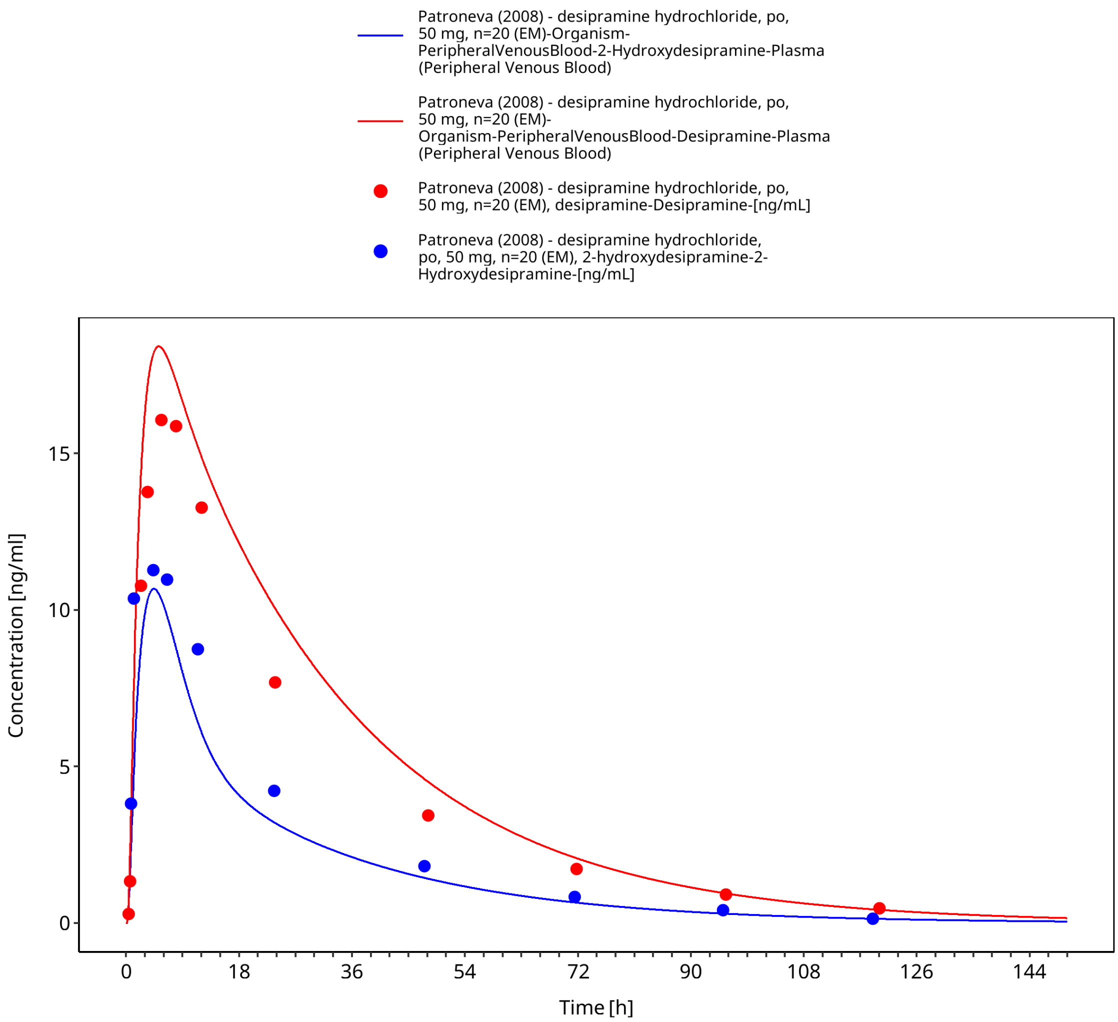

Figure 3-22: Time Profile Analysis

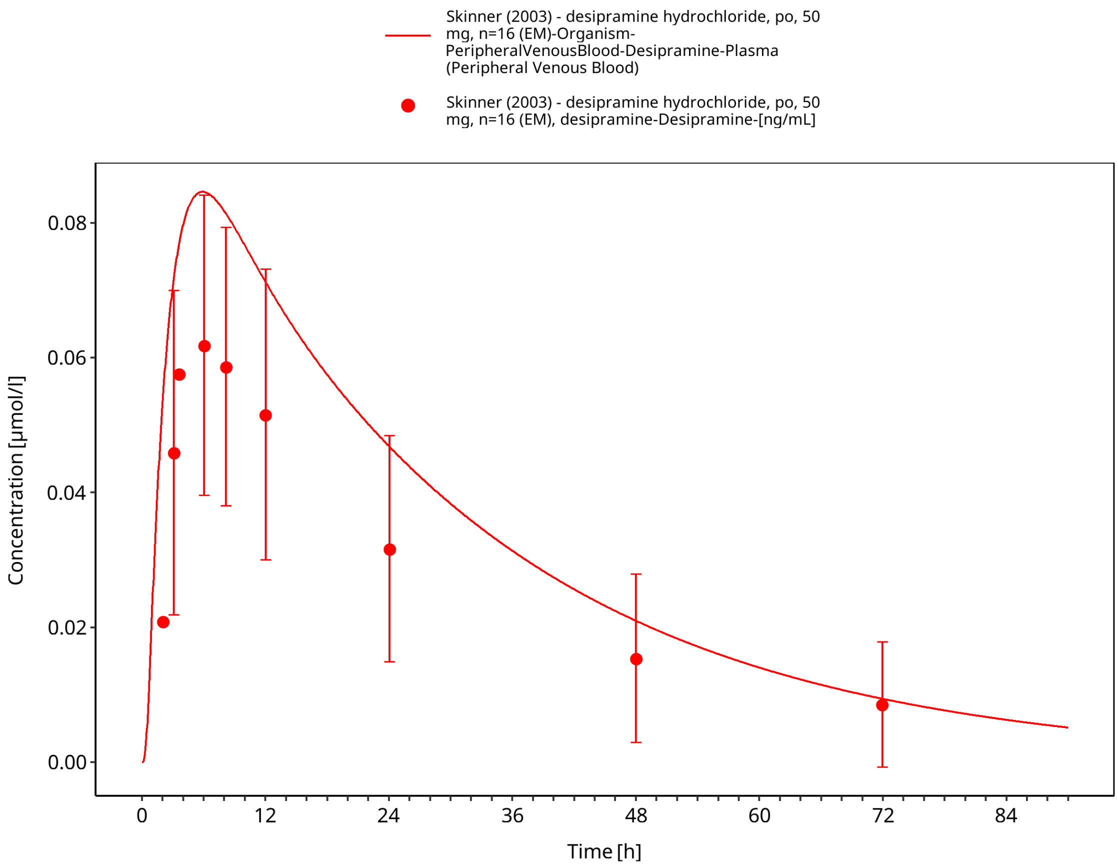

Figure 3-23: Time Profile Analysis

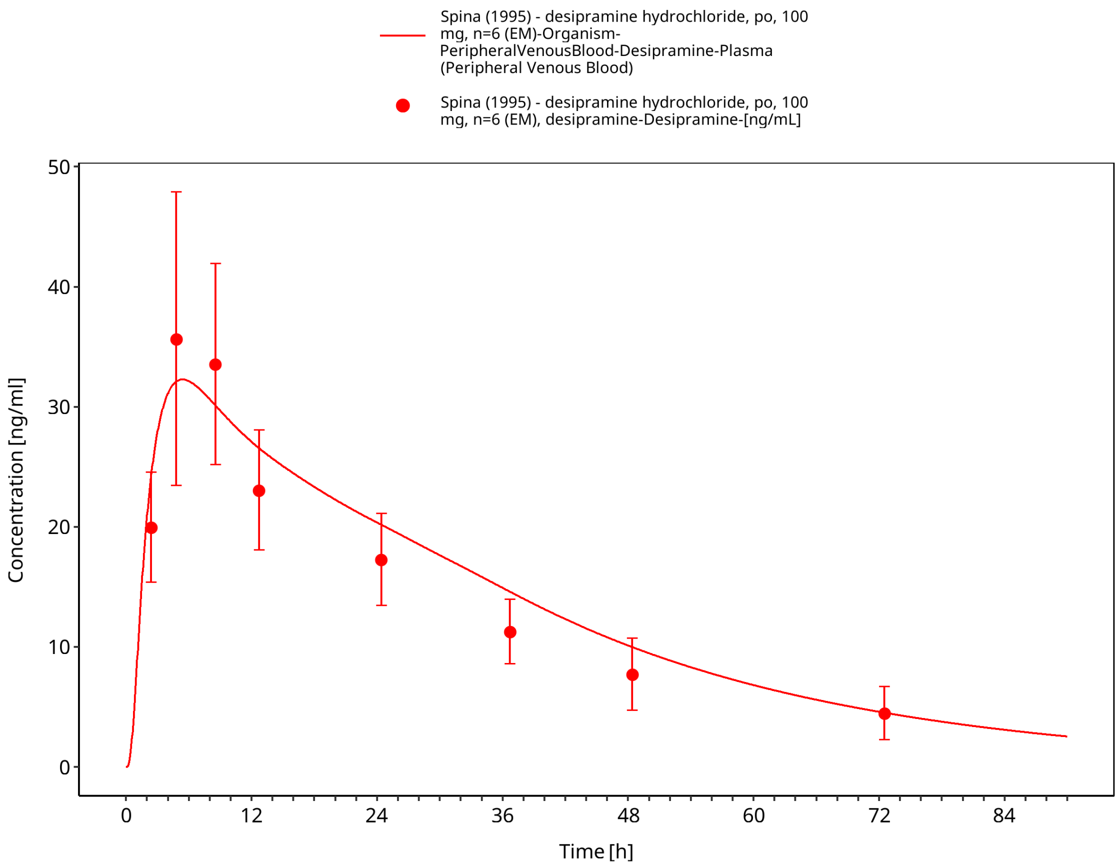

Figure 3-24: Time Profile Analysis

4 Conclusion

The presented PBPK model adequately describes the intravenous and oral pharmacokinetics of desipramine and 2-hydroxydesipramine in the evaluated adult CYP2D6 phenotype and activity-score groups.

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